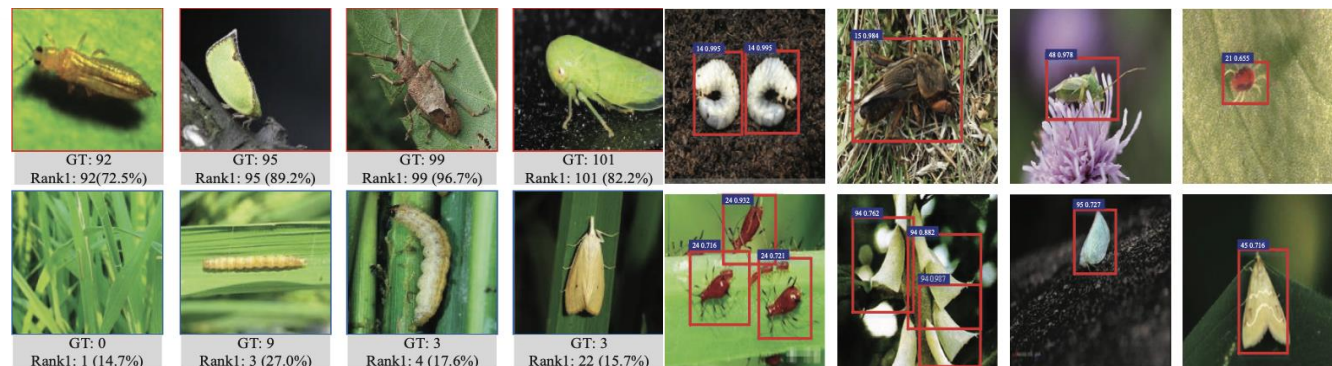


Insect pest specifies identification



BugCatchers

Meixuan Li (z5676001)

Xin Ye (z5665516)

Xinwen Xu (z5605611)

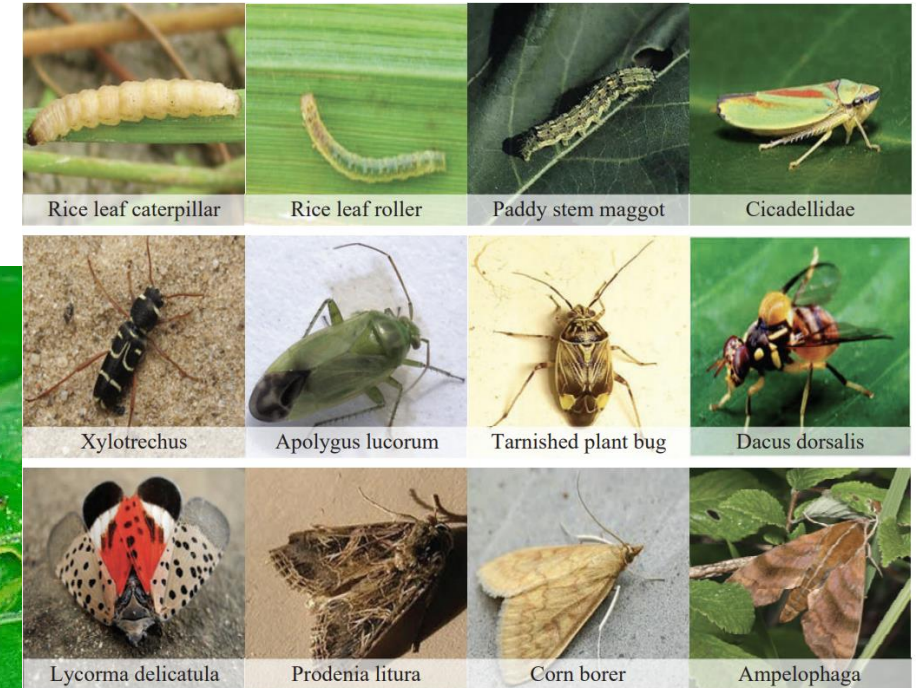
Yutian Wei (z5606653)

Thomas / Jian Shao (z5556885)

Motivation

Applications

- 🌱 Precision agriculture
- 🐛 Pest monitoring and early warning
- 🌱 Crop protection
- 🤖 Agricultural robots
- 📷 Automated field surveillance
- 🌍 Ecological monitoring



Problem Statement

Goal

- 102-class insect classification

Challenges

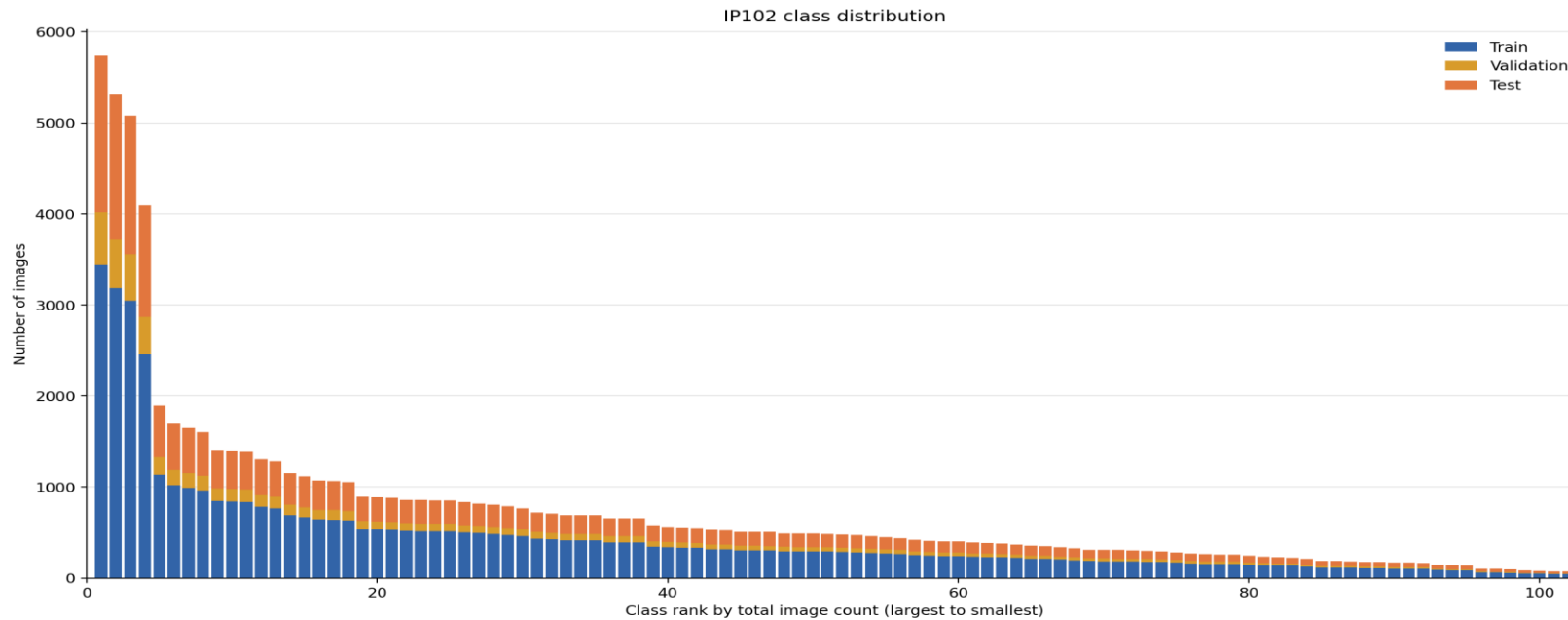
- 🔍 Fine-grained classification
- 🌿 Complex backgrounds
- 🌞 Variations in conditions
- ⚖️ Class imbalance



Dataset(s)

Classification Dataset

- 102 fine-grained insect pest species categories
- 75,222 images in total, before augmentation
- Long-tailed (highly imbalanced) distribution
- Split into 3 categories: train, val, test
- Largest class: Cicadellidae (5740 images)
- Smallest class: Erythroneura apicalis (71 images)



Source: local IP102 train.txt, val.txt, and test.txt split files; 102 classes.

Data Analysis

Train set

- 45,095 in total, max/min imbalance ratio: 82.00:1

Top 3 Most Frequent Labels

Label	Species	Images
101	Cicadellidae	3,444
67	Lycorma delicatula	3,186
70	Miridae	3,048

Top 3 Least Frequent Labels

Label	Species	Images
61	Brevipoalpus lewisi McGregor	47
80	Parlatoria zizyphus Lucas	44
72	Erythroneura apicalis	42

Data Analysis

Val set

- 7,508 in total, max/min imbalance ratio: 81.86:1

Top 3 Most Frequent Labels

Label	Species	Images
101	Cicadellidae	573
67	Lycorma delicatula	530
70	Miridae	508

Top 3 Least Frequent Labels

Label	Species	Images
63	Polyphagotars onemus latus	8
80	Parlatoria zizyphus Lucus	7
72	Erythroneura apicalis	7

Data Analysis

Test set

- 22,619 in total, max/min imbalance ratio: 78.32:1

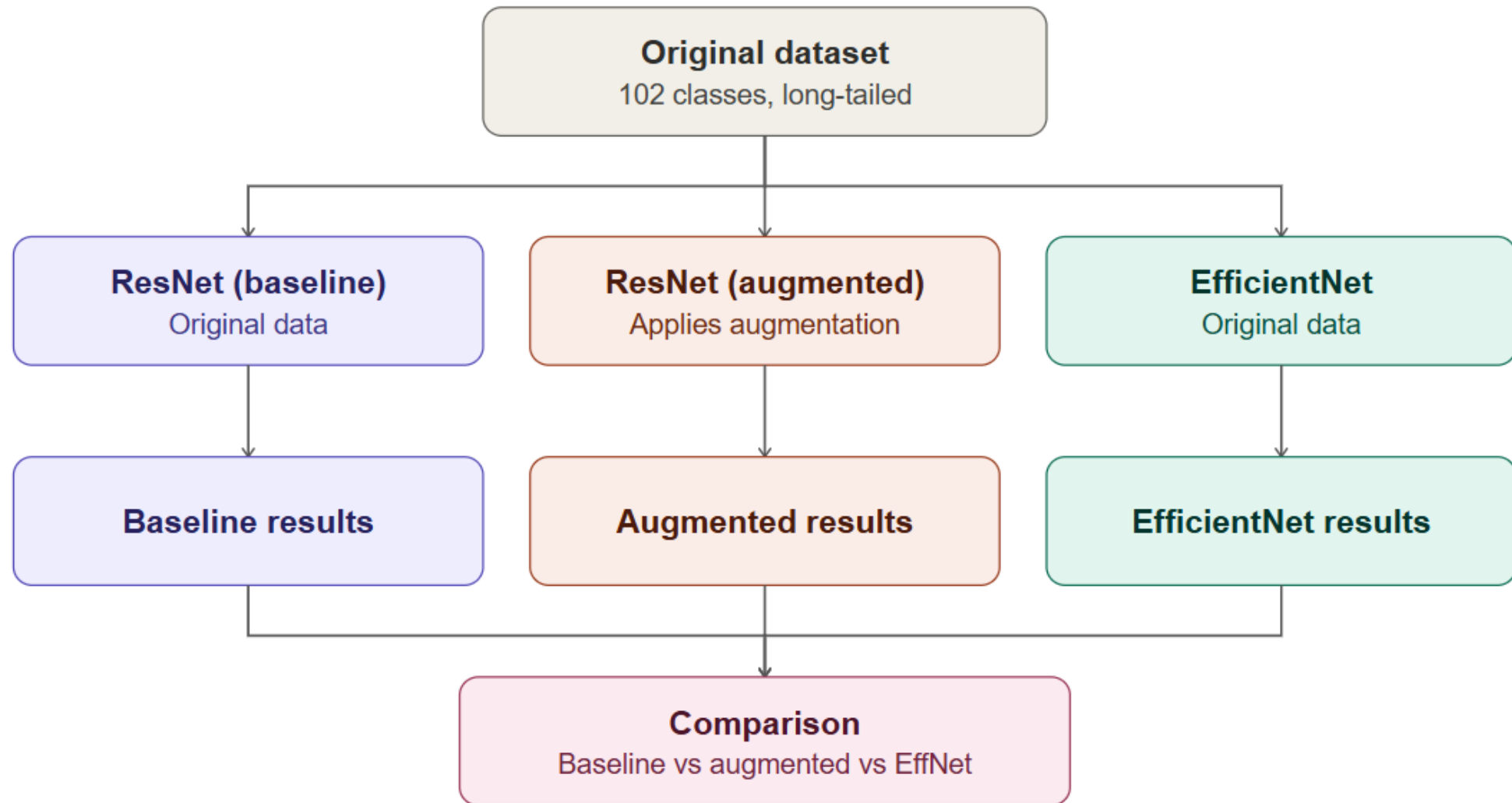
Top 3 Most Frequent Labels

Label	Species	Images
101	Cicadellidae	1,723
67	Lycorma delicatula	1,594
70	Miridae	1,525

Top 3 Least Frequent Labels

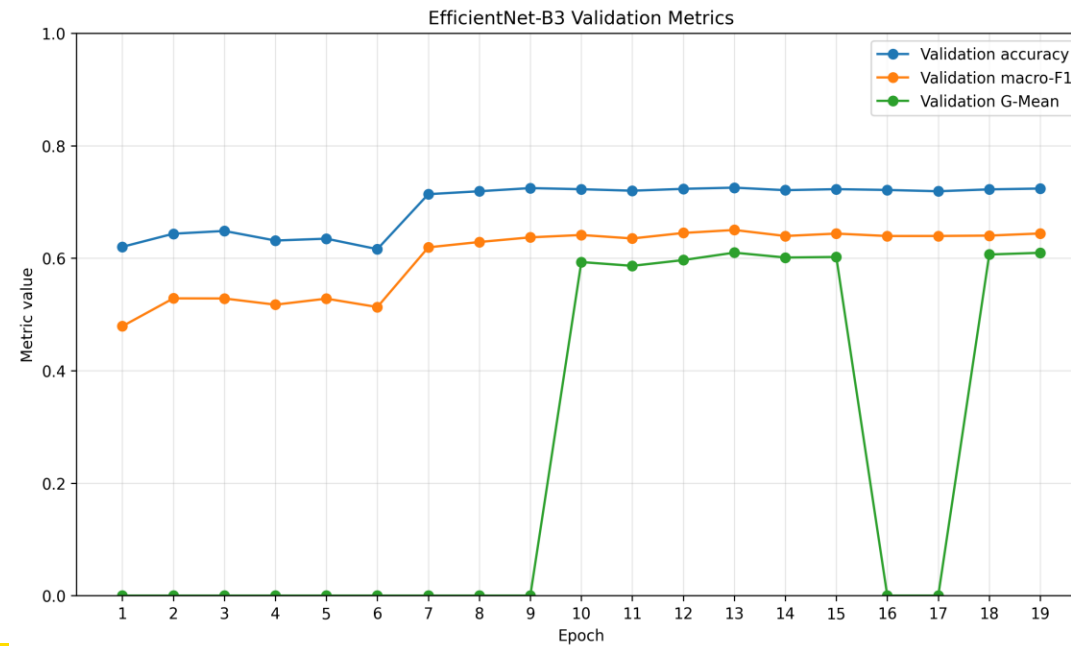
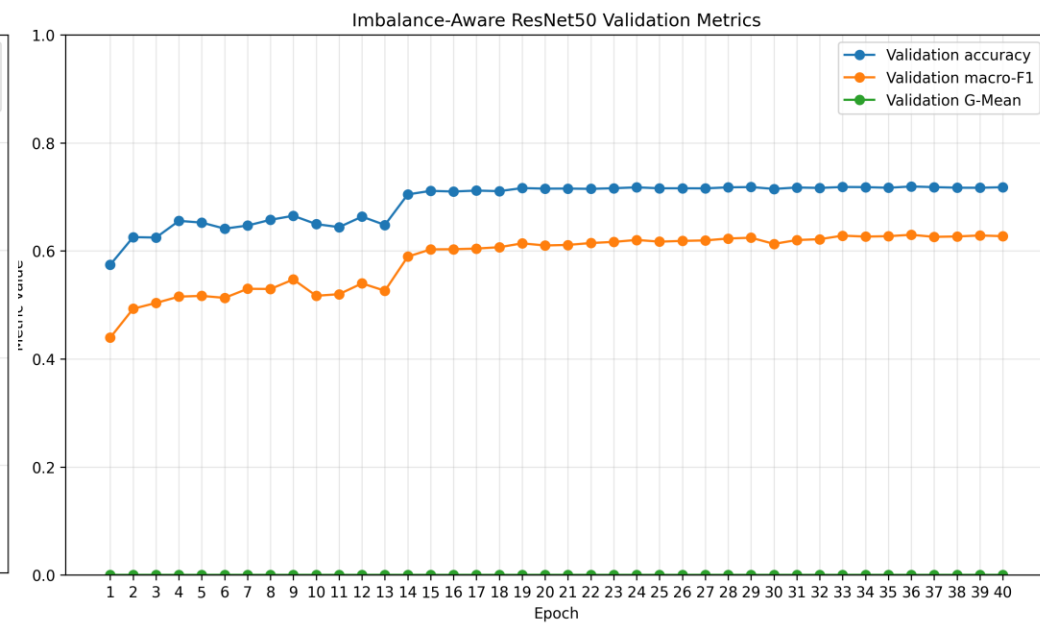
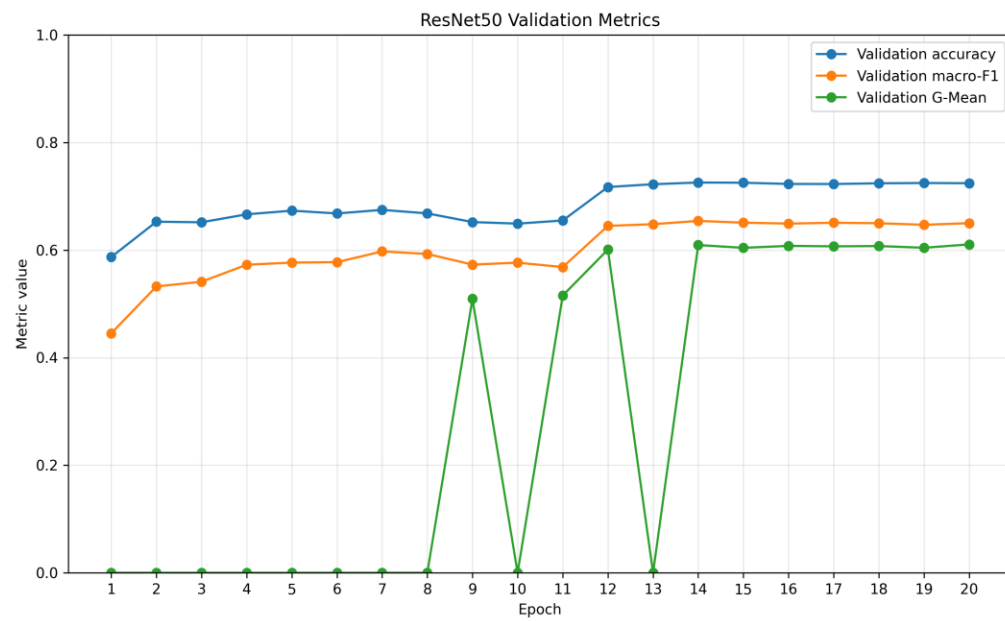
Label	Species	Images
61	Brevipoalpus lewisi McGregor	24
80	Parlatoria zizyphus Lucas	23
72	Erythroneura apicalis	22

Method(s)



Results

Validation Performance



Results(ResNet50 & Augmentation ResNet50)

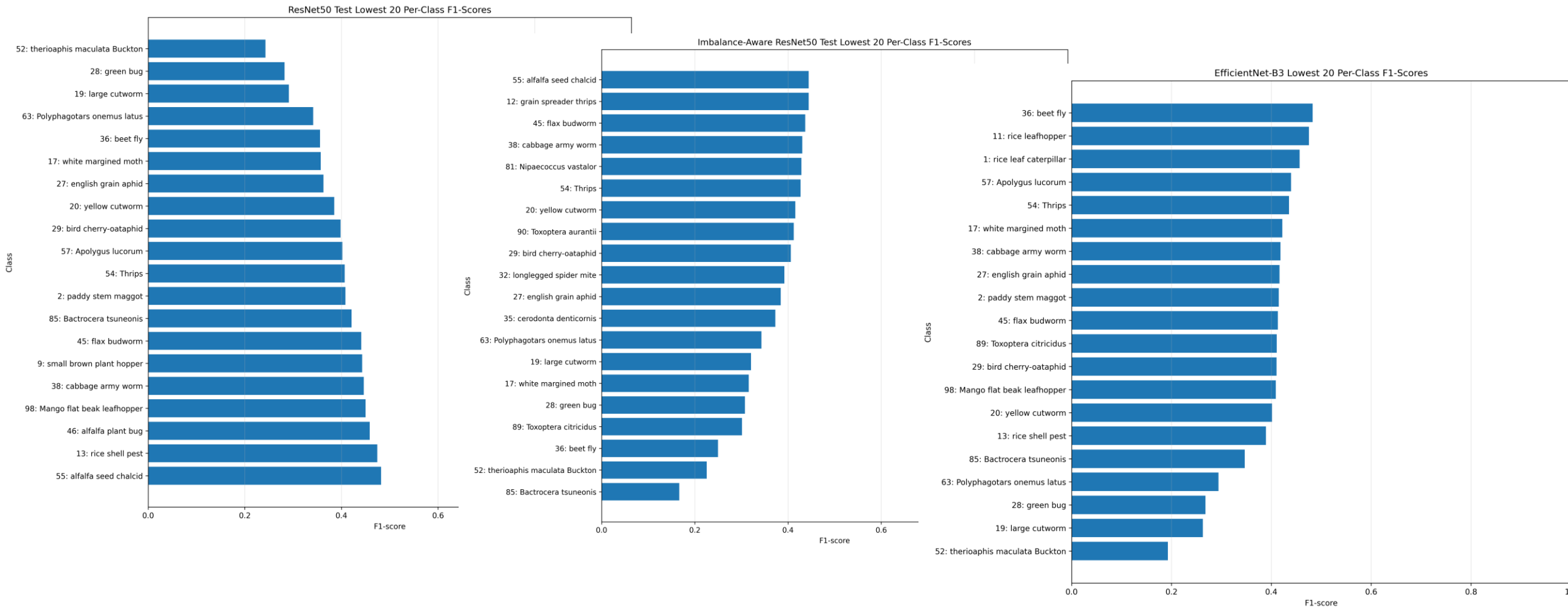
label	insect_name	precision	recall	f1-score	support
0	rice leaf roller	0.6904	0.8119	0.7462	335
1	rice leaf caterpillar	0.5328	0.4966	0.5141	147
2	paddy stem maggot	0.4412	0.3797	0.4082	79
3	asiatic rice borer	0.5888	0.5981	0.5934	316
4	yellow rice borer	0.6424	0.6382	0.6403	152
5	rice gall midge	0.7551	0.7303	0.7425	152
6	Rice Stemfly	0.6018	0.6126	0.6071	111
7	brown plant hopper	0.5531	0.4980	0.5241	251
8	white backed plant hopper	0.5000	0.5075	0.5037	268
9	small brown plant hopper	0.4528	0.4337	0.4431	166

insect_name	precision	recall	f1-score	support
rice leaf roller	0.6591	0.7791	0.7141	335
rice leaf caterpillar	0.5481	0.5034	0.5248	147
paddy stem maggot	0.5625	0.4557	0.5035	79
asiatic rice borer	0.5752	0.6171	0.5954	316
yellow rice borer	0.6376	0.6250	0.6312	152
rice gall midge	0.7244	0.7434	0.7338	152
Rice Stemfly	0.6210	0.6937	0.6553	111
brown plant hopper	0.5727	0.5179	0.5439	251
white backed plant hopper	0.5336	0.5634	0.5481	268
small brown plant hopper	0.5217	0.5060	0.5138	166

Results(Efficientnet)

label	insect_name	precision	recall	f1-score	support
0	rice leaf roller	0.6659	0.8209	0.7353	335
1	rice leaf caterpillar	0.4181	0.5034	0.4568	147
2	paddy stem maggot	0.5000	0.3544	0.4148	79
3	asiatic rice borer	0.5662	0.5823	0.5741	316
4	yellow rice borer	0.6036	0.6711	0.6355	152
5	rice gall midge	0.6444	0.7632	0.6988	152
6	Rice Stemfly	0.6897	0.5405	0.6061	111
7	brown plant hopper	0.5540	0.4701	0.5086	251
8	white backed plant hopper	0.5016	0.5821	0.5389	268
9	small brown plant hopper	0.4740	0.4940	0.4838	166

Results(Lowest 20 Per-Class F1-Scores)



Discussion1

- For Baseline resnet50's per-class metric(f1) on test-set, we want to find:
- On which insect classes resnet50 performs good and why?
- On which insect classes resnet50 performs bad and why?
- Define:

F1-score range	Number of classes	Interpretation
(F1 < 0.50)	?	Poor performance
(0.50 < F1 < 0.70)	?	Moderate performance
(0.70 < F1 < 0.85)	?	Good performance
(F1 > 0.85)	?	Excellent performance

Discussion1

Baseline ResNet50 Per-Class F1-score Distribution

F1-score range	Number of classes	Interpretation
$F1 < 0.50$	22	Poor performance
$0.50 \leq F1 < 0.70$	38	Moderate performance
$0.70 \leq F1 < 0.85$	29	Good performance
$F1 \geq 0.85$	13	Excellent performance

Discussion1

F1 < 0.50 (22 classes) -- Poor performance

No.	label	insect_name	precision	recall	f1-score	support
0	52	therioaphis maculata Buckton	0.2742	0.2179	0.2429	78
1	28	green bug	0.3333	0.2449	0.2824	98
2	19	large cutworm	0.2857	0.2973	0.2914	148
...	...	...	...	...	...	...
20	35	cerodonta denticornis	0.5429	0.4524	0.4935	42
21	89	Toxoptera citricidus	0.5	0.4912	0.4956	57

Discussion1

Is minority class always hard to classify?

No.	label	insect_name	precision	recall	f1-score	support
8	29	bird cherry-oataphid	0.3884	0.4085	0.3982	213
10	54	Thrips	0.4495	0.3712	0.4066	264
39	39	beet army worm	0.5771	0.6058	0.5911	482
73	24	aphids	0.7405	0.7917	0.7652	1229
87	78	Ceroplastes rubens	0.825	0.8462	0.8354	78
88	61	Brevipoalpus lewisi McGregor	0.9048	0.7917	0.8444	24
100	73	Papilio xuthus	0.9764	0.9185	0.9466	135
101	62	oides decempunctata	0.9524	0.9412	0.9467	85

Average support number in $F1 < 0.50$ (Poor performance): **124**

Average support number in $0.50 \leq F1 < 0.70$ (Moderate performance) : **159**

Discussion2

- For Baseline resnet50's confusion matrix on val-set, we want to find:
- 1. Which insect classes are similar and easily got misclassified with each other
- & 2. How to find them and group into a cluster?
- Graph theory: Automatically find / group similar nodes (to evaluate how similar two nodes are and stored this similarity as their weight and then grouping) as subgraph / community / cluster

Discussion2

Visual ConfusionScore = 100 * (0.40 * Cohesion + 0.30 * Severity + 0.20 * Reciprocity + 0.10 * Evidence Strength = 1)

cluster_name	number_classes	class_members	within_cluster_errors	cohesion	severity	reciprocity	visual_confusion_score
cluster_001	3	legume blister beetle blister beetle lytta polita	83	0.66936	0.21013	0.89157	60.9093
cluster_002	3	black cutworm large cutworm yellow cutworm	61	0.64211	0.33517	0.7541	60.8211
cluster_003	3	brown plant hopper white backed plant hopper ...	74	0.62712	0.32314	0.78378	60.4547
cluster_004	2	red spider Panonchus citri McGregor	24	0.64865	0.26374	0.66667	57.1914
cluster_005	3	english grain aphid green bug bird-cherry-oataphid	47	0.45631	0.27976	0.80851	52.8155
cluster_006	3	Miridae alfalfa plant bug tarnished plant bug	81	0.54	0.12348	0.81482	51.6006
cluster_007	4	army worm cabbage army worm beet army worm ...	94	0.46535	0.19502	0.80851	50.6347

Discussion2

Original (102) vs After merged (88)

split	evaluation_level	number_classes	number_samples	number_correct	number_errors	accuracy	macro_precision	macro_recall	macro_f1	g_mean
test	original	102	22619	16260	6359	0.72	0.66	0.64	0.65	0.60
test	merged	88	22619	17643	4976	0.78	0.71	0.67	0.69	0.65

Notes: (1) Classes in merged = $102 - 21 + 7 = 88$

(2) Merged evaluation doesn't mean model performance increases, the classification labels reduces from identifying 102 to 88, which may make task easier.

Discussion2

5 Difficulty level group's performance in Validation

split	difficulty_group	number_classes	number_samples	conditional_accuracy	macro_precision	macro_recall	macro_f1
val	cluster_hard	21	2203	0.6237	0.54683	0.52834	0.53407
val	non_cluster_total	81	5305	0.76814	0.70151	0.67777	0.68573



val	easy	37	3933	0.843377	0.825779	0.82637	0.82503
val	diffuse_hard	24	777	0.462033	0.505829	0.44136	0.46551
val	uncertain	20	595	0.670588	0.706433	0.68654	0.69229

Discussion2

Given 5 group's difficulty definition :

Group	Definition rule (based on validation results)
cluster_hard	A class belonging to an automatically discovered validation cluster retained by the visual-confusion threshold.
non_cluster_total	All classes outside the frozen discovered clusters; it is the union of the easy, diffuse_hard, and uncertain groups and serves as the comparison group for cluster_hard.
easy	Not in a selected confusion cluster; support ≥ 20 , F1-score ≥ 0.70 , recall ≥ 0.70 , and the most frequent wrong prediction accounts for less than 15% of the class support.
diffuse_hard	Not in a selected confusion cluster, but F1-score < 0.60 or recall < 0.60 ; the class performs poorly, but its errors are not concentrated within a selected cluster.
uncertain	Does not meet any of the above criteria; typically a class with intermediate performance, insufficient evidence, or low support.

Future work

split	difficulty_group	number_classes	number_samples	conditional_accuracy	macro_precision	macro_recall	macro_f1
val	cluster_hard	21	2203	0.6237	0.54683	0.52834	0.53407
val	diffuse_hard	24	777	0.462033	0.505829	0.44136	0.46551

- Common situation? Check minority class's metric & Verify cluster_hard & diffuse_hard's macro f1-score on all classifier models.
- More research on cluster_hard & diffuse_hard 's corresponding insect classes and their image quality.

Q&A

